Technical Project Assignment

Problem Statement

You have recently joined RetailMart Pvt. Ltd. as a Junior Data Engineer. The company runs a chain of retail stores across India and collects daily sales data from all its stores. The data is raw, messy, and stored in multiple CSV files. Your task is to build a small data pipeline that cleans, transforms, and loads this data so the business team can use it for reporting.

Scenario

RetailMart collects data from 3 sources every day:

- sales_data.csv — Contains daily sales transactions (may have missing values, duplicates, and wrong data types)
- products.csv — Contains product details like product name, category, and price
- stores.csv — Contains store information like store name, city, and region

The business team wants a clean, merged dataset that shows:

- Which products are selling the most in which city
- Total revenue generated per store per day
- Any stores or products with missing or incorrect data

Sample Data Structure

Before starting the tasks, create the following CSV files with at least 10-15 rows of sample data each. Make sure to include some missing values and duplicate rows in sales_data.csv to simulate real-world data issues.

sales_data.csv

Columns: sale_id, store_id, product_id, quantity, sale_date, amount

Note: Intentionally leave some rows with missing values in quantity and amount, and add a few duplicate rows.

products.csv

Columns: product_id, product_name, category, price

stores.csv

Columns: store_id, store_name, city, region

Tasks

Task 1: Data Ingestion

1. Load all three CSV files (sales_data.csv, products.csv, stores.csv) into pandas DataFrames. Print the shape and first 5 rows of each.

2. Check for missing values in all three DataFrames and print a summary of which columns have null values.

Task 2: Data Cleaning

3. Remove all duplicate rows from sales_data. Print how many duplicates were found and removed.

4. Fill missing values in the 'quantity' column with 0 and drop rows where 'amount' is NULL. Print the cleaned DataFrame shape.

5. Convert the 'sale_date' column to proper datetime format and the 'amount' column to float data type.

Task 3: Data Transformation

6. Merge all three DataFrames into one final DataFrame using appropriate JOIN on store_id and product_id. Print the final merged DataFrame.

7. Add a new column 'total_revenue' = quantity × price. Use NumPy to calculate and print the mean, max, and min of total_revenue.

8. Group the data by 'city' and find the total revenue generated per city. Sort the result in descending order.

Task 4: Data Loading (SQL)

9. Load the final cleaned and merged DataFrame into a SQLite database table called 'retail_sales'. Write the Python code using sqlite3 or sqlalchemy.

10. Write a SQL query to find the Top 3 best-selling products by total quantity sold from the 'retail_sales' table.

Task 5: Reporting & Insights

11. Write a SQL query to find total revenue per store per day from the 'retail_sales' table.

12. Using Python, print a summary report showing: Total number of transactions, Total revenue, Top selling city, and Top selling product.

Task 6: Pipeline & Error Handling

13. Write a Python function called run_pipeline() that runs all the above steps (load → clean → transform → load to DB) in one single function call.

14. Add basic error handling (try-except) to your pipeline so that if any file is missing, it prints a proper error message instead of crashing.

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